StudDFT keywords
(short manual, v1.08.3)

Program capabilities (v1.08.3)

- **Electronic-structure methods.** Restricted and unrestricted HF; MP2/UMP2 with spin-projected PUMP2 (energies, analytical gradients, FD-of-gradient Hessians); DFT: LDA (SVWN, SVWN3), GGA (BLYP, BP86, BPW91, PBE), global hybrids (B3LYP/B3LYP3, PBE0), meta-GGA (r2SCAN), and range-separated hybrids (wB97, wB97X, wB97X-D with built-in CHG dispersion, tunable via %rs_omega). Open-shell systems automatically switch to the U-prefixed variants.
- **Job types.** Single-point energies; analytical gradients; geometry optimization (redundant-internals RFO or Cartesian BFGS) with frozen bond/angle/dihedral constraints; harmonic frequencies with analytical Hessians (HF/DFT, incl. closed-shell range-separated hybrids and pure-functional RI-J) or semi-numerical gradient-FD Hessians, IR intensities on the analytical route; opt+freq; Boys–Bernardi counterpoise (BSSE); rigid and relaxed PES scans over Z-matrix coordinates; transition-state search (P-RFO) with mode following and .hess restart; TS+freq; intrinsic reaction coordinate (IRC) following via the Gonzalez–Schlegel GS2 method.
- **Excited states (TDDFT/TDHF).** Linear-response TDDFT/TDHF (Casida) for closed-shell HF/DFT references; full RPA or Tamm–Dancoff; singlet/triplet/both; oscillator strengths; transition and unrelaxed state dipole moments with a Generalized Mulliken–Hush coupling table (%tddft_dipmom); MOLDEN + CI-amplitude export for MultiWfn/THEODORE (%export_tddft); LR-PCM solvent response with nonequilibrium or equilibrium solvation.
- **Implicit solvation (PCM).** IEFPCM (default), C-PCM, and COSMO with named-solvent table or explicit permittivity; single points, analytical PCM gradients, optimizations, and semi-numerical PCM frequencies for HF and all DFT classes (RKS and UKS); LR-PCM coupling for TDDFT.
- **Empirical dispersion.** D3(BJ) and D4(BJ) corrections (energies, gradients, Hessians) for HF and common functionals; the Chai–Head-Gordon CHG tail is built into wB97X-D and enabled automatically.
- **Constrained DFT.** Wu–Van Voorhis charge-constrained DFT on Becke atomic fragments (single-point energies; restricted or unrestricted reference; multiple simultaneous constraints; two-loop Newton solver with analytic response Jacobian); 1e-integral and MO export tailored for diabatic coupling (S_AB, H_AB) construction.
- **Metallic and difficult SCF.** Fractional-occupation (Mermin) smearing — Fermi–Dirac, Gaussian, Methfessel–Paxton (N=1), and Marzari–Vanderbilt cold smearing — with fixed- or relaxed-moment UKS conventions; self-tuning damping + DIIS stabiliser for smeared metallic SCFs; AUTO level shift; SAD, core-Hamiltonian, or checkpoint initial guesses; MOM/Aufbau occupation control; SCF checkpoint / warm-restart machinery (%chkfile, %chk_write, %chk_every, %initial_guess read, %geo read).
- **Performance.** RI-J and RI-JK density fitting (def2-universal JFIT/JKFIT auxiliary sets; JFIT extended to K–Rn for heavy-element clusters in v1.08.3) with analytical RI gradients and an analytic RI-J Hessian for pure closed-shell functionals; direct and incremental (delta-P) SCF with an ERI-store memory cap; per-quartet auto HRR dispatch in the ERI kernels; optional multigrid SCF; OpenMP parallelism across the whole hot path (1e integrals, JK, XC quadrature, gradients, CPHF/CPKS).
- **ECPs.** Effective core potentials (built-in LANL2DZ, per-atom MIXED assignment) with numerical (Mura–Knowles quadrature) or analytical (McMurchie–Davidson) integral evaluation, including gradients and Hessians.
- **Analysis and output.** Mulliken charges plus extended population analysis (Löwdin, Hirshfeld, CM5 charges; Mayer/Mulliken/Löwdin bond orders; Mayer valences; spin populations); point-group detection with irreducible-representation labels for MOs and vibrational modes; static dipole polarizability from field CPKS; finite external electric field; MOLDEN orbital and normal-mode export; .hess files; 1e-integral print/export; per-timer profiling.

– **Basis sets.** Cartesian (default) or spherical-harmonic shells; built-in Pople and Dunning sets from STO-3G to 6-311++G(3df,2pd) and (aug-)cc-pVDZ/TZ/QZ plus LANL2DZ; per-atom MIXED basis assignment.

Keyword reference

In the table below, alternatives are separated by "│" and the default option is shown in square brackets or marked "default". Keywords, options and synonyms are case-insensitive (file names preserve case). Version tags (vX.YY.Z) mark the release in which a feature or its current behaviour was introduced.

Keyword	Options	Synonyms	Designation
Quantum chemical theories			
%method			Required. Electronic-structure method (case-insensitive).
	HF		Hartree–Fock
	MP2		HF reference + MP2 correlation (post-SCF); spin-projected PUMP2 for open shells (see %pump2)
	SVWN	SVWN5, LDA, LSDA	Slater + VWN-V LDA (NWChem / PySCF convention)
	SVWN3		Slater + VWN-III RPA (Gaussian 09 convention)
	BLYP		Becke88 + LYP GGA
	BP86	BP	Becke88 + Perdew86 GGA
	BPW91		Becke88 + Perdew–Wang 91 GGA
	PBE		Perdew–Burke–Ernzerhof GGA
	PBE0	PBE1PBE	PBE with 25 % HF exchange
	B3LYP	B3LYP5	B3LYP with VWN-V correlation (ORCA / NWChem)
	B3LYP3		B3LYP with VWN-III correlation (Gaussian convention)
	R2SCAN		r2SCAN meta-GGA (v0.79+). Depends on rho, |grad rho|^2 and tau.
	wB97		Range-separated hybrid, 100 % long-range HF exchange (alpha=0, beta=1, omega=0.4) (v0.88+)
	wB97X		Range-separated hybrid with 15.77 % short-range HF exchange (omega=0.3) (v0.88+)
	wB97X-D	wB97XD	wB97X re-fit with built-in Chai–Head-Gordon damped C6/R^6 dispersion (omega=0.2); the CHG tail is enabled automatically (v0.88.1+)
			Note: open-shell calculations automatically use U-prefixed variants (UHF, UMP2, UB3LYP, ...). RS functionals support energy / gradient / opt / scan / ts / irc / numerical freq; analytic RS Hessian and TDDFT are closed-shell (RKS) only.
%rs_omega	<positive real>		Override of the range-separation parameter omega (bohr^-1) for the wB97 family ("omega-tuning"; retunes both the exchange splitting and the SR-DFT attenuation) (v0.88.0). Error if the method is not range-separated.
Job type			
%job			Task to run. Default: energy.
	energy	sp, single	Single-point energy (default)

Keyword	**Options**	**Synonyms**	**Designation**
	gradient	grad, force	Analytical gradient at fixed geometry
	opt	optimize	Geometry optimization (internals RFO or Cartesian BFGS, see %opt_coord)
	freq	frequency, hessian	Hessian + harmonic vibrational frequencies; writes <base>.hess and <base>.molden
	opt+freq	optfreq	Opt followed by freq at the optimized geometry
	bsse	counterpoise, cp	Boys–Bernardi counterpoise (requires fragment IDs as a 5th column on every geometry line; fragments must be numbered contiguously from 1)
	scan	pescan, rigid_scan	Rigid PES scan; requires %gz block and %scan ... *end block
	scanopt	relaxed_scan, scan+opt	Relaxed PES scan: at every grid point a constrained geometry optimization is performed
	ts	opt_ts, optts	Transition-state search via P-RFO (HF / DFT only)
	ts+freq	tsfreq, opt_ts+freq, opttsfreq	TS search followed by freq pass at located TS; writes <base>_ts.hess / <base>_ts.molden
	irc		Intrinsic-reaction-coordinate walk (Gonzalez–Schlegel GS2) from a stored TS Hessian; requires %read_hess (v0.86.0). Gas-phase HF / DFT only (no MP2 / PCM / CDFT). Writes <base>_irc_forward.xyz / <base>_irc_reverse.xyz trajectories.
Molecule builder			
%charge	<signed integer>		Total molecular charge. Default: 0.
%mult	<positive integer>	%multiplicity	Spin multiplicity 2S+1 (1 = singlet, 2 = doublet, ...). Default: 1.
%spin	restricted │ unrestricted	RHF, RKS, R │ UHF, UKS, U	SCF spin treatment. If not set, determined automatically from %mult (open shells switch to unrestricted).
%geo	[angstrom │ bohr] │ read	%geometry, *geometry │ read: restart	Start of Cartesian geometry block. Terminated by *end or empty line. Atom token: <Z> │ <Element> │ <Element><index>. "%geo read" (v1.08.2) takes the element list and coordinates from the checkpoint file (see %chkfile) instead of the block; any atom lines still present are parsed and discarded with a NOTE.
%gz			Start of Z-matrix geometry block (Angstroems, degrees). Terminated by empty line. Required for %job scan / scanopt.
Basis set			
%basis	<basis set name>		Required. Basis set name (case-insensitive). Built-in: sto-3g, 3-21g, 6-31g, 6-31g(d), 6-31g(d,p), 6-31+g(d), 6-31++g(d,p), 6-311g(d), 6-311g(d,p), 6-311g(2d,2p), 6-311g(3df,2pd), 6-311++g(2d,2p), 6-311++g(3df,2pd), cc-pVDZ, cc-pVTZ, cc-pVQZ, aug-cc-pVDZ, aug-cc-pVTZ, aug-cc-pVQZ, LANL2DZ. Pople star forms are resolved automatically (6-31G* = 6-31G(d), 6-31G** = 6-31G(d,p)). MIXED: per-atom assignment via <selector> <basis_name> lines until next %-keyword or empty line.
%spherical	(flag)		Use spherical-harmonic d/f-shells (5d/7f); default is Cartesian (6d/10f).

Keyword	**Options**	**Synonyms**	**Designation**
ECP			
%ecp	<ecp name>		Effective core potential. Built-in: LANL2DZ. MIXED: per-atom assignment via <selector> <ecp_name> lines until next %-keyword or empty line.
%ecp_method	[numerical] │ analytical	numerical: quadrature, num, 0 │ analytical: md, mcmurchie, 1	ECP integral evaluation method. Default: numerical (Mura–Knowles radial quadrature). Analytical = McMurchie–Davidson (matches g09 to ~1 microHa on AlH3-class molecules since v0.77.1). For very tight basis primitives (alpha > ~5000) prefer numerical.
Electronic integral calculations			
%boys_cook	(flag)	%boys_precise	Use Cook recurrence for Boys function instead of fast tabulated representation. Default: off (table method).
%hrr_mode	[auto] │ contracted │ primitive	contracted: new, fast │ primitive: old, legacy	HRR placement in the HGP ERI kernels (v0.91–0.95). AUTO (default) picks the cheaper kernel per shell quartet (contraction-first iff the quartet contraction product Kq >= 2). "contracted" / "primitive" force one path; primitive reproduces historical references bit-exactly.
SCF procedure			
%scf_converger	[DIIS] │ NONE	%converger	Convergence accelerator. Default: DIIS. Internally tests whether the value starts with "DIIS"; anything else turns DIIS off.
%scf_maxiter	<positive integer>		Max number of SCF iterations. Default: 150. Also propagated to the SAD atomic SCFs (effective atomic budget max(50, %scf_maxiter), v1.07.2).
%scf_etol	<positive real>		Energy convergence threshold (Hartree). Default: 1e-8.
%scf_ptol	<positive real>		Density-matrix convergence threshold. Default: 1e-6.
%scf_diis_etol	<positive real>	%scf_gtol	DIIS error-norm (orbital-gradient) convergence threshold (v0.76.0). All three (%scf_etol, %scf_ptol, this one) must be satisfied simultaneously. Default: 1e-5.
%scf_min_iter	<positive integer>		Minimum SCF iterations before convergence can be declared (v0.76.0). Default: 5.
%scf_n_converged	<positive integer>	%scf_nconv	Consecutive iterations all three convergence criteria must be satisfied (v0.76.0). Default: 2.
%scf_diis_cond_max	<real > 1>	%scf_diis_condmax	Upper bound on DIIS B-matrix condition number; oldest vector is pruned when exceeded (v0.76.0). Default: 1e10.
%damping	<real in [0,1]>		Density-matrix damping factor. DIIS overrides damping after a few iterations. Default: 0.0. (Smeared metallic SCFs use their own self-tuning damping, v1.07.4.)
%level_shift	AUTO │ OFF │ <non-negative real>		Virtual-orbital level shift (Hartree). Useful for problematic open-shell / heavy-element cases. Default: AUTO (turning on shift 0.2 if SCF is not converged in 100 steps). Forced OFF under %smearing.
%initial_guess	AUTO │ SAD │ CORE │ READ	%guess, %scf_guess; HCORE, CORE_H; READ: chk, restart	Initial-guess strategy (v0.76.0). Default: AUTO (RHF/RKS -> CORE; UHF/UKS -> SAD if a fit basis is loadable, CORE otherwise). SAD supports ECP atoms. READ (v1.08.0) seeds the SCF from the checkpoint file (%chkfile); a failed read is fatal by design. Cross-spin/-multiplicity reads convert correctly

Keyword	**Options**	**Synonyms**	**Designation**
			(warm restarts for sigma-ladders and multiplicity scans).
%uks_occupation	AUTO │ STRICT │ MOM │ MIX	%uhf_occupation, %occupation; AUFBAU, MAXOV, MAX_OVERLAP, REMIX	UHF/UKS orbital-occupation strategy (v0.76.0). Default: AUTO (MOM for the first iterations, then strict Aufbau). No effect for restricted; bypassed under %smearing.
%eri_memory_limit_mb	<positive integer, MB>	%eri_mem_limit_mb, %eri_memory_limit, %eri_mem_limit	ERI-buffer memory cap (v0.76.3); falls back to direct JK when exceeded. Default: 4096.
%scf_direct	on │ [off]		Force the direct SCF: skip the in-memory ERI store and rebuild integrals every iteration (memory-bound regime of large molecules) (v0.90.2). Default: off.
%scf_incremental	[on] │ off		Incremental (delta-P) Fock builds in the direct-SCF path with Schwarz x max|dP| screening; full rebuild every 20 iterations (v0.90.2). Default: on (in direct mode).
%scf_multigrid	off │ on │ <preset 1..10>	%multigrid	Multigrid SCF (trial mode, v0.99.0): converge on a coarse DFT grid first, then finish on the target %grid from that density. Final energy is converged on the target grid. %job energy only; ignored for HF and CDFT. Default: off.
SCF checkpoint / warm restart (v1.08.0+)			
%chkfile	<filename>	%chk_file	Checkpoint file name (case preserved). Default: input file name with the extension replaced by ".chk". The checkpoint stores density matrices (RHF: half density; UHF: Pa, Pb), spin treatment, nbf, natom, charge, multiplicity, element list, geometry (bohr) and the converged energy in a plain-text, portable format.
%chk_write	[on] │ off		Write the checkpoint after every successful molecular SCF (each converged optimizer step overwrites it, so after a run it holds the LAST geometry and density). Default: on.
%chk_every	<integer >= 0>		Also write the checkpoint every n iterations DURING the SCF (crash / power-loss recovery for long metallic-cluster runs) (v1.08.3). Default: 0 (converged-only).
			Restart recipe: "%geo read" restores the geometry and "%initial_guess read" restores the density from the checkpoint. Validation on read: hard fail on nbf / natom / element mismatch; warn on charge, multiplicity, geometry.
Fractional-occupation smearing (finite electronic temperature, v1.07.0+)			
%smearing	fermi │ gauss │ mp │ cold │ [off]	%smear; fermi: fd │ gauss: gaussian │ mp: mp1, methfessel-paxton │ cold: mv, marzari-vanderbilt	Occupation-smearing scheme for the Mermin (finite-T) SCF; HF and DFT, RHF/RKS and UHF/UKS. Default: off. The SCF converges on and reports the Mermin free energy F = E - T*S; the final block prints the bare E, -T*S, the sigma->0 estimate E0 ~ (E+F)/2, the Fermi level(s) and the fractional occupations. Gates: %job energy only; no MP2, %tddft or %cdft. Level shift and MOM are bypassed; SAD atomic SCFs run unsmeared; PCM rides along.
%smearing_sigma	<positive real, Ha>	%smear_sigma, %smearing_width	Smearing width sigma in Hartree (for Fermi–Dirac sigma = kB*T_el). Default: 0.01.
%smearing_moment	[fixed] │ relaxed	fixed: nupdown │	UKS moment convention under smearing (v1.07.1).

Keyword	Options	Synonyms	Designation
		relaxed: relax, shared, common, auto, free	fixed: two chemical potentials conserve N_alpha and N_beta separately (multiplicity preserved; VASP NUPDOWN analogue). relaxed: one shared mu conserves only N_total; the magnetization relaxes to the free-energy optimum (the input multiplicity only seeds the guess). No effect for RKS/RHF.
Density fitting (RI, v0.96.0+)			
%ri	[off] │ j │ jk	j: on, rij │ jk: rijk, k	Resolution-of-the-identity (density-fitting) Coulomb / exchange builds. "j": fitted Coulomb for PURE functionals (SVWN, BLYP, BP86, BPW91, PBE, r2SCAN), RKS and UKS; the four-index ERI phase is skipped entirely. "jk" (v0.98.0): fitted J and K, opens HF and global hybrids. Jobs: energy / gradient / opt / freq / opt+freq (analytical RI gradients, v0.97–0.98.3); the analytic Hessian covers pure closed-shell functionals with %ri j (v1.02.0), other cases use the semi-numerical route. Not with range-separated functionals, MP2, %tddft, %cdft, scan / ts / irc / bsse. Auxiliary bases loaded automatically: def2-universal-JFIT (H–Ar; extended, decontracted K–Rn in v1.08.3) for "j", def2-universal-JKFIT (H–Ar) for "jk". PCM rides along. Default: off.
%ri_check	on │ [off]		One-shot fitted-vs-exact J diagnostic on the first SCF iteration (prints max|dJ|, dE_J). Validation instrument; costs about as much as a full conventional 2e phase — keep off in production runs.
DFT options			
%grid		%grid_quality	DFT integration-grid quality. Default: 3 (standard). No effect for HF or MP2.
	1		Coarse (25 radial, Mura–Knowles, Lebedev up to 110, no pruning)
	2		Medium (50 radial, 3-tier pruning, up to 302)
	3		Standard (75 radial, 3-tier pruning, up to 590; default)
	4		Extended (100 radial, up to 590)
	5		Extreme (150 radial, up to 590)
	fine	gfine	Preset 6: Gaussian Fine / NWChem (75 radial Euler–Maclaurin, NWChem 5-zone pruning up to 302, Becke size adjustment)
	ultrafine	gultrafine	Preset 7: Gaussian UltraFine (99 radial, NWChem 5-zone pruning up to 590)
	tm	treutler	Preset 8: Treutler–Ahlrichs M4 (75, 302) uniform 302-point angular, no pruning (PySCF-like)
	g09	g09ref	Preset 9: G09 reference (75, 302) uniform 302-point angular, Euler–Maclaurin + Becke size adjustment
	orca7		Preset 10: ORCA Grid-7 reproduction (Treutler–Ahlrichs M3 radial, element-dependent radial count via Krack–Koester, 5-zone pruning {110, 434, 590, 770, 590} with genuine Lebedev-770; v0.78.2/0.81.6)
%xc_deriv	[analytical] │ numerical	analytical: anal, exact │ numerical: num, fd	Analytical or finite-difference XC functional derivatives during SCF and DFT gradient / Hessian. Default: analytical.

Keyword	Options	Synonyms	Designation
Dispersion			
%dispersion	NONE │ D3 │ D4	%disp; D3: D3BJ, D3-BJ, D3(BJ) │ D4: any name containing "D4"	Empirical dispersion correction (energy, gradient, Hessian). Default: NONE. D3(BJ) parameters built in for: HF, BLYP, BP86/BP, BPW91, B3LYP*, PBE, PBE0. D4(BJ) (v0.83+) parameters built in for: HF, BLYP, BP86/BP, B3LYP, PBE, PBE0. D3(zero) is not implemented. Cannot be combined with the wB97 family — use %method wB97X-D, whose built-in CHG tail is switched on automatically.
MP2 options			
%frozen_core	-1 │ 0 │ <positive integer>	%nfrozen	Frozen-core control. -1 (default) = auto-detect noble-gas core of every non-ECP atom; 0 = correlate all; N = freeze lowest N occupied orbitals (per spin for UMP2).
%pump2	[yes] │ no	on, true, 1, T, <bare> │ off, false, 0, F	Spin-projected open-shell MP2 (Schlegel-1986 single-annihilation). Default: yes — PUMP2 single-point diagnostic always printed when UMP2 is active. With %pump2 yes, FD-PUMP2 optimization is also enabled (gradient w.r.t. PUMP2 energy). With %pump2 no, PUMP2 is fully suppressed.
Geometry optimization			
%geom_maxiter	<positive integer>	%opt_maxiter, %opt_max_iter, %geom_max_iter	Max optimization steps. Default: 50.
%grad_tol	<positive real>		Maximum-component gradient convergence threshold (Ha/Bohr). Default: 4.5e-4.
%grad_method	[analytical] │ semianalytical	analytical: full, 1 │ semianalytical: semi, fd, 0	Gradient method. PUMP2 gradient is always numerical (h = 1e-3 Bohr).
%opt_coord	[internal] │ cart	%opt_coordinates, %opt_coords │ internal: internals, redundant, rfo, int │ cart: cartesian, bfgs, cart_bfgs, cartesian_bfgs, xyz	Optimizer coordinate system (v0.75.0). Default: internal (redundant-internals RFO). Cartesian path does NOT support constraints or PCM. Diatomics always use Cartesian BFGS.
%initial_hess	[diag] │ full	%initial_hessian, %init_hess, %model_hess, %guess_hess; diag: diagonal, lindh │ full: almlof, almloef, coupled	Initial model Hessian for the RFO/internal-coordinate minimizer (v0.81.0). diag (default): Lindh diagonal model. full: coupled Almloef/Lindh model with off-diagonal force constants (helps molecules with strongly coupled modes).
Geometry constraints (partial-constraint optimization)			
%constrain			Start a constraint block (v0.70+). One declaration per line, terminated by *end or empty line. Compatible with %job opt / opt+freq, internal-coordinate optimizer only.
	B <i> <j>		Freeze bond between atoms i and j
	A <i> <j> <k>		Freeze angle i-j-k (vertex at j)
	D <i> <j> <k> <l>		Freeze dihedral i-j-k-l
PES scan (rigid / relaxed)			

Keyword	Options	Synonyms	Designation
%scan			Start a sweep-variable block (v0.69+). Required by %job scan and %job scanopt. One sweep variable per line, terminated by *end or empty line. Requires a %gz Z-matrix geometry.
	<kind> <row> [from] <val_from> [to] <val_to> [step] <val_step>		<kind> = R (bond) │ A (angle) │ D (dihedral); <row> = 1-based Z-matrix row index. Keywords from/to/step are optional.
Hessian / Frequencies			
%hess_method	[analytical] │ numerical	analytical: full, 1 │ numerical: num, fd, 0	Hessian computation method. Default: analytical (HF, DFT incl. ECP; closed-shell range-separated hybrids since v0.88.2; %ri j with pure closed-shell functionals since v1.02.0; static polarizability piggybacks, see %polar). Numerical = semi-numerical finite difference on the analytical gradient (h = 5e-3 Bohr); it is auto-selected (with a NOTE) for MP2/UMP2, PCM, open-shell RS functionals, and the remaining RI cases.
TS options (transition-state search)			
%read_hess	<filename>	%read_hessian	.hess file to load before TS search (v0.72.0) — and REQUIRED by %job irc (v0.86.0). Reader verifies geometry, method, basis, ECP, charge, multiplicity, spin, format version (for IRC the file geometry is authoritative and coordinates are deliberately not compared). If omitted for a TS search, the TS driver computes the Hessian itself at the input geometry.
%ts_mode	<positive integer>	%tsmode	1-based mode index to follow on the first P-RFO step (v0.72.0). After step 1, optimizer switches to maximum-overlap mode tracking. Default: 1 (lowest mode).
IRC options (reaction-path following, v0.86.0+)			
%irc_step	<real in (0, 2]>		Path step size in amu^1/2*bohr (GS2 pivot at s/2, corrector sphere radius s/2). Default: 0.10.
%irc_maxpoints	<integer in [1, 10000]>	%irc_max_points	Max path points per branch. Default: 25.
%irc_gtol	<positive real>		Termination threshold on max |dE/dx| (Ha/bohr): "minimum region reached". Default: 4.5e-4 (same scale as %grad_tol).
%irc_direction	forward │ reverse │ [both]	%irc_dir; forward: f │ reverse: backward, r, b	Which branch(es) of the path to follow from the TS. Default: both.
%irc_mode	<positive integer>		Which imaginary mode to start along (1-based, by ascending Hessian eigenvalue). Default: 1 (correct for every true TS).
TDDFT / TDHF excited states (v0.87.0+)			
%tddft	<nstates, 1..1000>		Compute <nstates> excited states by linear-response TDDFT / TDHF (Casida) after the SCF. Requires %job energy and a closed-shell (RHF/RKS) reference; not with MP2, %cdft, %smearing or meta-GGA (r2SCAN) kernels. PCM is supported via the LR-PCM response kernel (v0.94.0). Output: state table (Ha / eV / nm / oscillator strength) + dominant amplitudes + RPA de-excitation diagnostics.

Keyword	Options	Synonyms	Designation
%tddft_tda	on │ [off]		Tamm–Dancoff approximation (CIS-like) instead of full RPA-type TDDFT. Default: off.
%tddft_multiplicity	[singlet] │ triplet │ both	%tddft_mult; s, t	Spin symmetry of the computed excitations. Default: singlet. Triplet oscillator strengths are 0 (spin-forbidden).
%tddft_dipmom	on │ [off]		Transition dipoles <0|mu|n> and <m|mu|n>, unrelaxed excited-state dipole moments, and a two-state Generalized Mulliken–Hush electron-transfer coupling table (v0.87.1). Default: off (cheap).
%export_tddft	on │ [off]	%tddft_export	Write <base>_tddft_mo.molden (ground-state MOs) and <base>_tddft_states.txt (Gaussian-layout excitation blocks with CI amplitudes) for MultiWfn / THEODORE hole–electron / NTO analyses (v0.87.1). Default: off.
%tddft_pcm	[nonequilibrium] │ equilibrium	noneq, neq │ eq	Solvation regime of the LR-PCM TDDFT kernel (v0.94.0). nonequilibrium (default): kernel at the optical permittivity eps_inf = n^2 (vertical absorption). equilibrium: kernel at the static eps (full solvent relaxation).
%eps_optical	<real >= 1>		Optical permittivity eps_inf for the nonequilibrium LR-PCM kernel. Resolution: explicit %eps_optical > built-in n^2 table (18 named solvents) > 2.0 fallback with a NOTE.
%tddft_selftest	on │ [off]		Runtime numerical self-tests of the kernel contraction machinery, incl. the (A+B)-column vs CPKS identity and the LR-PCM response check (teaching / debugging aid). Default: off.
PCM (implicit solvent, v0.73.0+)			
%pcm	none │ cpcm │ iefpcm │ cosmo	cpcm: c-pcm, c_pcm │ iefpcm: ief-pcm, ief_pcm, pcm │ none: off, no, false, 0, F	Continuum solvation model. Default: none. %solvent or %eps auto-activate IEFPCM if no %pcm is given (v0.74.0; matches Gaussian SCRF=PCM — note this changed from the earlier C-PCM auto-default). Supported with PCM: %job energy, gradient, opt (analytical PCM gradients, v0.74.1), freq, opt+freq (semi-numerical PCM Hessian, v0.93.0), and %tddft (LR-PCM, v0.94.0). RKS and UKS; all functional classes incl. range-separated (v0.92.0). Not with MP2 or %job bsse / scan / ts / irc.
%solvent	<name>		Named solvent (resolved to tabulated permittivity; overrideable by %eps). Recognized: water (h2o), methanol (meoh), ethanol (etoh), acetone, acetonitrile (mecn), dmso, dmf, thf, chcl3 (chloroform), ccl4, dichloromethane (dcm, ch2cl2), toluene, benzene, cyclohexane, hexane (n-hexane), diethylether (et2o, ether), ethyl_acetate (etoac), ammonia (nh3), vacuum (none).
%eps	<real >= 1.0>	%epsilon, %pcm_eps	Direct override of dielectric permittivity. Vacuum = 1.0.
%pcm_lebedev	<positive integer>		Per-atom Lebedev quadrature order on cavity surface. Default: 110. Allowed: 6, 14, 26, 38, 50, 74, 86, 110, 146, 170, 194, 230, 266, 302, 350, 434, 590, 770.
%pcm_scaling	<positive real>		Scaling factor on Bondi vdW radii for atomic-sphere cavity. Default: 1.20 (standard COSMO/CPCM).
%pcm_switch	<positive real, Bohr>		Width of cubic-taper switching function on cavity-

Keyword	Options	Synonyms	Designation
			tessera occupancy. Default: 0.20 Bohr (typical 0.1 - 0.4).
%pcm_grad_method	[analytical] │ numerical	%pcm_gradient_method │ analytical: ana, an, 1, true, yes, on │ numerical: num, fd, 0, false, no, off	PCM gradient method (v0.74.x). Default: analytical (fast). Numerical = central-difference FD (6N+1 PCM SCFs), ~10-30x slower; agrees with analytical to ~1e-5 a.u./bohr; retained for cross-validation.
Constrained DFT (v0.84.0+)			
%cdft	[becke]		Start a charge-constrained DFT block (Wu–Van Voorhis; Becke atomic weights are the only scheme in this release). One fragment definition per line; the block ends at a blank line, *end, or the next %-keyword. %job energy only (gradients / optimization are a later stage); not with %tddft, %smearing, %ri or range-separated functionals. The reported energy is the physical E_KS at the constrained density; converged Lagrange multipliers and CDFT orbital energies are printed.
	<label> <atom indices...> charge <q_target>		Fragment definition line, e.g. "Donor 1 2 3 charge +1.0". Label is free text; atom indices are 1-based; the literal keyword "charge" separates the atom list from the target fragment charge. Multiple independent fragments are handled through the full Newton system (do not let two fragments cover ALL atoms — the Jacobian becomes singular).
%cdft_tol	<positive real>		Outer-loop tolerance on max |N_c - N_target| (electrons). Default: 1e-4.
%cdft_maxiter	<positive integer>		Max outer (Lagrange-multiplier) Newton iterations. Default: 50.
%cdft_spin	[auto] │ restricted │ unrestricted	r, rks, rhf │ u, uks, uhf	Spin treatment of the CDFT diabat (v0.84.1). auto: RKS for a closed-shell singlet (robust), UKS otherwise. restricted forces RKS (error if open-shell); unrestricted forces UKS (e.g. broken-symmetry biradical diabats; the historical v0.84.0 behaviour).
%cdft_level_shift	<non-negative real, Ha>	%cdft_levshift	Floor for the Roothaan level shift of the inner constrained SCF (pure stabiliser: converged density / energy / Jacobian unchanged). Effective shift = max(%level_shift, %cdft_level_shift). Default: 0.2 (RKS) / 0.5 (UKS); crank up (1.5–2.0) for hard open-shell diabats; 0 disables.
Properties and analysis			
%pop_analysis	[on] │ off	%population, %pop	Extended population analysis after the always-on Mulliken charges (v1.05.0): Löwdin charges; Hirshfeld charges (grid integration against the program's own spherically averaged SAD promolecule) + CM5 charges (parametrized for Z <= 20, warning above); Mayer / Mulliken (overlap) / Löwdin (Wiberg) bond orders; Mayer bonded / free / total valences; Mulliken, Löwdin and Hirshfeld spin populations for UHF. Bare keyword = on; default (absent): off.
%symmetry	[on] │ off	%sym, %pointgroup, %point_group	Point-group symmetry analysis (v1.06.0): detects the molecular point group and labels canonical MOs and vibrational normal modes by irreducible representation; prints the electronic configuration and the Gamma_vib summary. Supported: C1, Cs, Ci,

Keyword	Options	Synonyms	Designation
			Cn/Cnv/Cnh/Sn/Dn/Dnd/Dnh (n <= 8), Td, Oh, and linear C∞v / D∞h; icosahedral and rarer cubic groups are analyzed in their largest supported axial subgroup with a printed note. The geometry is never reoriented or symmetrized — all energies are unchanged. Default (absent): off.
%symmetry_tol	<positive real, bohr>	%sym_tol, %symtol	Atom-matching distance tolerance for detecting a symmetry operation. Default: 1e-3.
%polar	[on] │ off	%polarizability	Static dipole polarizability from a 3-RHS field CPKS solve, computed alongside the analytic frequencies (%job freq / opt+freq on the analytical-Hessian route) (v1.04.0). Closed shell only (UHF/UKS prints a NOTE and skips). Prints the Gaussian "Dipole polarizability, Alpha" block (iso, aniso, lower triangle; au / esu / SI). Works with %ri j and hybrids. Default (absent): off.
%efield	<fx> <fy> <fz> (au)	%field	Uniform external electric field added to H_core before the SCF (finite-field properties; also the FD oracle for %polar) (v1.04.0). The dipole print then reports mu(F). Default: 0 0 0.
Output control			
%print_mo	(flag)		Print converged MO coefficient matrix at end of SCF.
%print_1e	NONE or <filename>	%print_1e_integrals	Print 1e matrices (S, T, Vnuc, Hcore) in the WORKING basis of the run (spherical if %spherical, else Cartesian) to stdout, or to <filename> if given (v0.84.1).
%print_time	(flag)	%print_timing	Print detailed timing breakdown per timer counter at end of run.
%print_grad	(flag)	%print_gradient	Print per-atom Cartesian gradient at every optimization step (format: <Element><index> gX gY gZ).
%export_mo	(flag)	%molden_mo, %export_orbitals	Save converged MO coefficient matrix in Molden format to the external file <base>_mo.molden (Cartesian convention, viewer-ready per-component normalization; v0.82.0). Written for every job type that ends with a single wavefunction (energy, gradient, opt, freq, optfreq, ts, ts+freq); for MP2 the canonical HF orbitals are exported.
%export_1e	(flag)	%export_1e_integrals	Save 1e matrices (S, T, Vnuc, Hcore) to <base>_1e.export in the SAME Cartesian basis, AO order and normalization as the %export_mo molden file, so C_A^T M C_B with molden coefficients is convention-consistent (intended for CDFT S_AB / H_AB coupling construction; v0.84.1).
Comments			
# or !			"#" at the start of a line: whole line is ignored. "!" anywhere on a line: the rest of the line is ignored — in-line comments ARE supported since v0.74.0 (e.g. "%job opt ! comment"). TAB, CR and other non-printable characters are replaced by spaces before parsing.